

AAAI 2027 Reproducibility Checklist — PHAROS

Paper Title: PHAROS: Navigated Hierarchical Attention with Adaptive Budgets, Cross-Level Guidance, Tidal Momentum, and Uncertainty-Aware Fallback

Submission Type: AAAI 2027 Main Technical Track

Authors: Anonymous Submission

Summary Statement

This paper introduces PHAROS, a method and theoretical framework for adaptive hierarchical attention in long-context transformers. The current submission does not report verified quantitative experimental results. Instead, it provides the method, mathematical formulation, theoretical properties, limitations, and a reproducible evaluation protocol. No unsupported numerical claims are made.

Checklist Answers

1. Does the paper include empirical experimental results?

Answer: No. The current submission does not report verified quantitative experimental results. The paper intentionally avoids unsupported numerical claims and presents an evaluation protocol to be executed in future verified runs.

2. Does the paper include theoretical claims?

Answer: Yes. The paper includes theoretical claims regarding cost-neutral budget allocation, selection smoothing through exponential moving averages, and uncertainty-triggered fallback computation.

3. Are theoretical claims accompanied by proof sketches or derivations?

Answer: Yes. The main paper includes concise theoretical discussion, and the supplementary material provides additional proof and complexity details.

4. Does the paper introduce a new dataset?

Answer: No. PHAROS does not introduce a new dataset.

5. Does the paper use existing datasets?

Answer: Not in the current empirical sense. The proposed evaluation protocol recommends using the same architecture family, tokenizer, dataset mixture, and training recipe as the Lighthouse Attention baseline to ensure fair comparison, but no dataset-based results are reported in the current submission.

6. Does the paper provide code?

Answer: No. Code is not included with this submission. The method is specified mathematically and algorithmically, including the four PHAROS mechanisms: Depth-Sounding Budget Allocation, Range Light Cross-Level Guidance, Tidal Momentum Selection, and Fog Gate Confidence Fallback.

7. Are model architecture and implementation details described?

Answer: Partially. The method is designed as a thin wrapper over Lighthouse-style hierarchical selection and efficient attention pipelines. The paper describes the mathematical formulation and algorithmic composition. Future empirical submissions should include exact implementation details, software stack, and hardware setup.

8. Are hyperparameters reported?

Answer: Partially. The paper defines mechanism-level hyperparameters such as entropy temperature, minimum head budget, cross-level guidance scale, momentum coefficient, and confidence threshold. Final tuned values are not reported because experiments have not yet been completed.

9. Are compute resources reported?

Answer: Not applicable for the current submission. Since the paper does not report experimental runs, GPU type, number of devices, training duration, and compute budget are not reported. These should be added after verified experiments.

10. Are random seeds and variance reported?

Answer: Not applicable. No quantitative experiments are reported. Future experiments should report random seeds, number of runs, and variance or confidence intervals where appropriate.

11. Are baselines specified?

Answer: Yes. The paper specifies recommended baselines: dense SDPA, Lighthouse Attention with norm scorer, Lighthouse Attention with dilated scorer, and PHAROS with individual mechanisms ablated.

12. Are evaluation metrics specified?

Answer: Yes. The proposed evaluation protocol includes training loss, resume loss, validation perplexity, long-context retrieval accuracy, throughput, memory usage, and wall-clock latency.

13. Are ablation studies specified?

Answer: Yes. The paper proposes ablating Depth-Sounding Budget Allocation, Range Light Cross-Level Guidance, Tidal Momentum Selection, and Fog Gate Confidence Fallback while keeping the rest of the training pipeline fixed.

14. Does the paper include limitations?

Answer: Yes. The main limitation is the absence of verified quantitative experiments in the current draft. The paper also notes that theoretical assumptions about entropy, recall curvature, and cross-layer selection stability require empirical validation.

15. Does the paper make unsupported empirical claims?

Answer: No. The paper explicitly avoids unsupported numerical claims and does not state empirical superiority over Lighthouse Attention, dense SDPA, or other baselines.

16. Does the paper include ethical or broader impact considerations?

Answer: Partially. The paper focuses on efficient long-context transformer training. Potential positive impact includes reduced computational overhead for long-context training. Potential risks include enabling larger-scale model training, which may increase downstream deployment risks if models are misused. Future versions should include a fuller broader-impact statement if required by the venue.

17. Are supplementary materials provided?

Answer: Yes. Supplementary material includes proof details, fog-gate analysis plan, and complexity derivation. Reviewers are not required to review supplementary material; therefore, all essential method details are included in the main paper.

18. Is the submission anonymous?

Answer: Yes. The paper is prepared as an anonymous submission and does not include author names, affiliations, acknowledgments, or self-identifying text.

19. Is the paper formatted according to AAAI requirements?

Answer: The LaTeX source is prepared to use the AAAI 2027 style files `aaai2027.sty` and `aaai2027.bst`. Authors should compile with the official AAAI 2027 Author Kit and verify the final PDF respects the page limit before submission.

20. Final reproducibility note

Answer: PHAROS is reproducible at the method-specification level in the current submission. Full empirical reproducibility requires future release of code, exact datasets, model configurations, training scripts, compute details, random seeds, and verified experimental results.